fails outside it. If you memorise answers to all the practise questions, you might do splendidly if those very questions come up, but fail otherwise.

The real-world concept of stereotypes encapsulates these concepts quite well. People learn to make assumptions about people based on their class / race / religion / gender. One can develop a stereotype of people of a particular type based on someone they know. This is like overfitting, because these stereotypes will not hold up in real life, they only apply to the person you know. On the other hand, if you learn a stereotype from others and try to base decision on that, your decision are based on poor and incomplete data.

Categories of Machine Learning

As we may have hinted before there are different categories of Machine Learning. There are different kinds of problems that one might try to apply machine learning to, and they define the approach we need to take towards building a model of the data.

Supervised Learning

We already mentioned supervised learning when we discussed spam filtering earlier. Spam filtering is essentially looking at a piece of data (an email with extended information about its headers) and deciding whether it fits in the spam category or not. This in itself does not imply that it is supervised learning, since it is an example of the application of what has been learned.

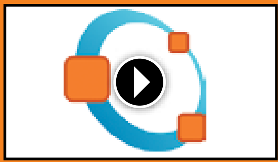
Supervised learning then is about how we train the computer to recognise spam. In this case we are feeding the algorithm a large number of emails that are already marked as spam or not spam. It is like teaching children what is hot and what is cold by giving them examples of both. In supervised learning, we provide data that is already labelled, which the computer trains on and then.

We can take this example further and look at what Google is doing now with Gmail; putting email into different categories (Social, Promotion, Updated, Forum, etc.). Here it isn't just a matter of spam or not, but of deciding which category an email belongs to. GMail can learn from how people apply these labels to their mails and do it automatically in the future.

While supervised learning makes it sound like a human needs to manually create, feed or categorise the data, this is not exactly true. The data can come from anywhere with the only condition being that it should be correctly labelled.

*pointers

>>Interesting developer videos



*Octave tutorials

>>Octave is a free and open source alternative to MATLAB by GNU, that uses a MATLAB-compatable syntax. This extensive video series on Octave teaches you how to use this software in depth. These skills are applicable in MATLAB as well.

http://dgit.in/1BQwamu

*Scikit-learn

>>This fourty-minute long talk delivered at the EuroPython 2014 Python convention takes a look at scikit-learn, an open source machine learning library for Python. Despite the short length it covers a lot.

http://dgit.in/ScikitLearn

*Introduction to R

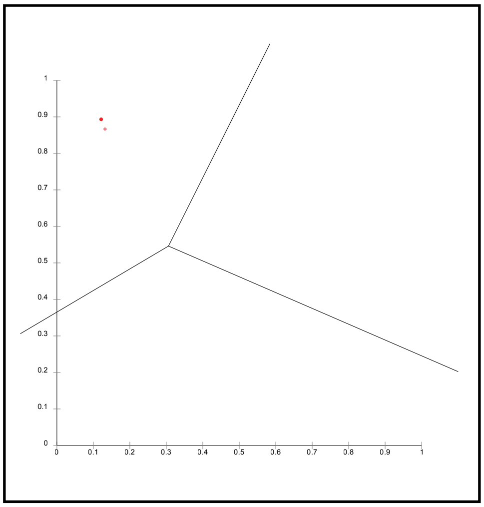
>>R is a great programming language for the mathematically oriented. It has a wealth of features when it comes to dealing with large amounts of data, such as filtering, processing, and visualisation.

http://dgit.in/RLangIntro

*Machine Learning in R

>>R is great at dealing with large amounts of data. This is a meaty lecture that also includes introductory content about machine learning itself.

http://dgit.in/MachineR



k-means clustering can divide data into clusters by starting with random clusters and repeatedly reevaluating the centre and membership of clusters.

Unsupervised Learning

We gave the example above of how children can learn what is hot and what is cold by giving them example of each. The fact is though, that a child could easily learn the difference between hot and cold without being explicitly told which is hot and which is cold. The child might not have the vocabulary to describe them as hot or cold, but could still put them in different categories in its mind. It's more likely that a child already thinks of things as hot and cold, and simply learns to apply those labels as and when he/she comes across new experiences.

Computers can similarly learn things from data, and put them into different categories without being explicitly told what kind of data belongs to which category.

For example, on a shopping website, the developers might not know in advance what kinds of users will visit it, and what kind of shopping habits they will have. An unsupervised learning algorithm could be used in such an instance to segregate the users into different segments and cater to their needs based on other users like them. This is one of the steps taken towards building a good recommendation engine.

Another great example of where this can be used in biology, in analysing gene sequences. It can be used to cluster gene sequences, and find relations between genes.

Think of any situation where data can be grouped into segments, whether you know those segments in advance or not, and you have a candidate for implementing unsupervised machine learning.